# Camille Hankel

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Cooperative Institute for Climate, Ocean, and Ecosystem Studies, University of Washington

Dept. of Atmospheric and Climate Science, University of Washington

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## Education

Ph.D., Earth and Planetary Sciences, Harvard University. May 2024. *Advisor: Eli Tziperman.*
Secondary Field in History of Science.

B.S., Mathematics, Georgetown University, 2018. *Summa cum laude.*

B.A., Computer Science, Georgetown University, 2018.

## Positions

Postdoctoral Fellow at the Cooperative Institute for Climate, Ocean, and Ecosystem Studies (CICOES) and the Dept. of Atmospheric and Climate Science at the University of Washington. *Sept. 2024–present*

## Teaching

Teaching Fellow, EPS 231: Climate Dynamics, Harvard University, 2023

Teaching Fellow, EPS 131: Introduction to Physical Oceanography and Climate, Harvard University, 2022

Head Teaching Fellow, APM120: Applied Linear Algebra and Big Data, Harvard University, 2020 & 2021

Course Development for EPS 101: Global Warming Science, Harvard University, 2020

In collaboration with my PhD advisor Eli Tziperman I helped develop one unit of this new course, designing the curriculum, generating the instructional text, coding excercises, and figures, and contributing to the publication of a textbook chapter based on the final course materials.

Teaching Assistant, Data Structures, Georgetown University, 2016

Teaching Assistant, Computer Science I & II, Georgetown University, 2015-2016

## Awards & Fellowships

NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellowship, 2024

CICOES Postdoctoral Fellowship at the University of Washington, 2024

National Center for Atmospheric Research ASP Postdoctoral Fellowship, 2024 (declined)

Cooperative Institute for Research in Environmental Sciences at the University of Colorado Boulder Postdoctoral Fellowship, 2024 (declined)

AGU Ocean Sciences Section Outstanding Student Presentation Award, Fall Meeting 2023

Recipient of NCAR's Exploratory Allocation Computing Grant, 2023

US CLIVAR Travel Grant to participate in the ENSO Summer School, Trieste, Italy, 2022

Bok Center Certificate of Distinction in Teaching, Harvard University, 2021

Earth and Planetary Sciences Teaching Award, Harvard University, 2020

Harvard Skaff Family Environmental Graduate Fellowship, 2018

Phi Beta Kappa Member, Georgetown University, 2018

Clare Booth Luce Undergraduate Scholarship, Georgetown University, 2016-2018

## Manuscripts in preparation/submitted

1. Zhang, Sally and Camille Hankel. "Millennial-Scale transient and equilibrium responses of regional eddy-driven jets to $CO_2$ doubling." *In prep.*

2. Bonan, David and Camille Hankel. "Forcing-rate dependent transient climate sensitivity." *In prep.*

3. Hankel, Camille, Edward Blanchard-Wrigglesworth, and Haruki Hirasawa. "Increasing boreal fire emissions suppress AMOC decline." *Submitted.*

4. Yu-Chiao Liang, David Bonan, Lily C. Hahn, Nicole Feldl, Camille Hankel, Ivy Tan, Lise Selan Graff, Marius Arthun, Doyeon Ki, Mark R. England, and Felix Pithan. "Arctic amplification in observations and models of varying complexity [review article]." *Under review at Nature Reviews Earth and the Environment.*

## Publications

1. Hankel, Camille, and David Bonan. "Transient evolution of polar amplification under different $CO_2$ ramping rates." Geophysical Research Letters 53 (2026): e2025GL120079. https://doi.org/10.1029/2025GL120079

2. Hankel, Camille, Wei Cheng, and Cecilia Bitz. "Arctic sea ice meltwater as a forcing and feedback on the Atlantic Meridional Overturning Circulation." Journal of Climate (2026): e250254. https://doi.org/10.1175/JCLI-D-25-0254.1

3. Hankel, Camille. "The effect of $CO_2$ ramping rate on the transient weakening of the Atlantic Meridional Overturning Circulation." *Proceedings of the National Academy of Sciences* 122.1 (2025): e2411357121. https://doi.org/10.1073/pnas.2411357121

4. Hankel, Camille, and Eli Tziperman. "Assessing the robustness of Arctic sea ice bi-stability in the presence of atmospheric feedbacks." *Journal of Geophysical Research: Atmospheres, 128* (2023), e2023JD039337. https://doi.org/10.1029/2023JD039337

5. Hankel, Camille, and Eli Tziperman. "An approach for projecting the timing of abrupt winter Arctic sea ice loss." *Nonlinear Processes in Geophysics* 30.3 (2023): 299-309. https://doi.org/10.5194/npg-30-299-2023

6. Hankel, Camille, and Eli Tziperman. "Chapter 2: Greenhouse." *Global Warming Science*, Princeton University Press, 2022.

7. Hankel, Camille, and Eli Tziperman. "The Role of Atmospheric Feedbacks in Abrupt Winter Arctic Sea Ice Loss in Future Warming Scenarios." *Journal of Climate* 34.11 (2021): 4435-4447. https://doi.org/10.1175/JCLI-D-20-0558.1

8. Kogay, Roman, et al. "Machine-learning classification suggests that many alphaproteobacterial prophages may instead be gene transfer agents." *Genome biology and evolution* 11.10 (2019): 2941-2953. https://doi.org/10.1093/gbe/evz206

## Invited Talks

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| December 2026 | Invited speaker for "Polar Climate Change and Variability" session, AGU Annual Meeting |
| October 2026 | Invited speaker for "Polar Ocean Circulations and their Climate Effects" session, iSPOGOC conference, Qingdao, China |
| May 2026 | CGD Seminar, National Center for Atmospheric Research, Boulder, CO |
| March 2026 | Ocean and Climate Physics Seminar, Lamont-Doherty Earth Observatory, Columbia University |
| December 2025 | Equilibrium Climate Sensitivity and Cloud Feedbacks Virtual Symposium |
| October 2025 | Caltech Engineering & Applied Sciences Trailblazers Symposium and Environmental Science and Engineering Department Seminar |
| August 2025 | Research Seminar, College of Earth, Ocean, and Atmospheric Sciences, Oregon State University |
| June 2025 | Interagency Arctic Research Policy Committee (IARPC) Physical Oceanography Webinar |
| April 2025 | Physical Oceanography Seminar, University of Washington, Seattle |
| January 2025 | Atmospheric and Climate Science Seminar, University of Washington, Seattle |
| October 2023 | Workshop on Non-autonomous Dynamics in Complex Systems, Max-Planck Institute for Physics of Complex Systems, Dresden, Germany |
| September 2023 | Max Planck Institute for Meteorology Joint Seminar, Hamburg, Germany |

## Presentations, Posters, and Workshops

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| May 2026 | "Collapse of the Atlantic Ocean Circulation: Can it? Has it? Will it?" workshop, London, UK |
| December 2019–2024 | American Geophysical Union Fall Meeting |
| September 2023 | Dynamics Days Europe |
| July 2022 | 3rd Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics, International Centre for Theoretical Physics in Trieste, Italy |
| June 2019 & 2022 | 22nd & 23rd Conference on Atmospheric and Oceanic Fluid Dynamics |
| November 2019 | 13th Graduate Climate Conference, Woods Hole Oceanographic Institute |

## Outreach & Service

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| *Present* | Research Mentor for recently-graduated Atmospheric Sciences student Steven Xu, University of Washington |
| 2024–2025 | Member of the Committee for Equity, Diversity, and Inclusion, Dept. of Atmospheric and Climate Science, University of Washington |
| 2025 | Volunteer guest speaker for ATM S 220, *Exploring the Atmospheric Sciences*, a general-interest undergraduate course at the University of Washington. |
| 2025 | Session Organizer and Co-Chair for "The State and Future of Polar Climate: Insights from Idealized Models, Reanalyses, and Projections", 18th Conference on Polar Meteorology and Oceanography at the 2025 AMS Denver Summit |
| 2024 | Session Organizer and Co-Chair for "History and Philosophy of the Geosciences: What's Next?", AGU Fall Meeting 2024 |
| 2022–present | Peer Reviewer for *Nature Climate Change*, *Nature Geoscience*, *PNAS*, *GRL*, *Journal of Climate*, *Journal of Physical Oceanography*, *npj Climate and Atmospheric Science*, *Environmental Research Letters*, *Earth System Dynamics*, and *Journal of Physics: Complexity* |
| 2023 | Volunteer for the Harvard EPS Graduate Admissions Application Program: an initiative that offers support and feedback to prospective PhD applicants from groups underrepresented in STEM. |
| 2020–2021 | Mentor for the Science Research Mentorship Program (SRMP): a program that aims to introduce local (Cambridge, MA) high-school students from diverse backgrounds to the Earth and Planetary Sciences through year-long research projects. |
| 2020–2022 | Member of the Diversity, Inclusion, and Belonging (DIB) History of Racism Subgroup: an effort to lead a reflection on and study of Harvard's own historical legacy of racism. |
| 2020 | Mentor for Harvard EPS Short-Term Summer Program: an initiative which aims to recruit undergraduate students from diverse backgrounds to the EPS major at Harvard by offering paid summer research positions. |